Faculdade de Ciências e Tecnologia da Universidade de Coimbra

ASSIGNMENT 2

Algoritmos de Diagnóstico e de Auto-Regulação
2019/2020

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1. Introduction

In this work we identified the discrete-time transfer function of the *Feedback's Process Trainer PT326* and developed a controller using MATLAB and Simulink. The data used were previously extracted from *PT326*.

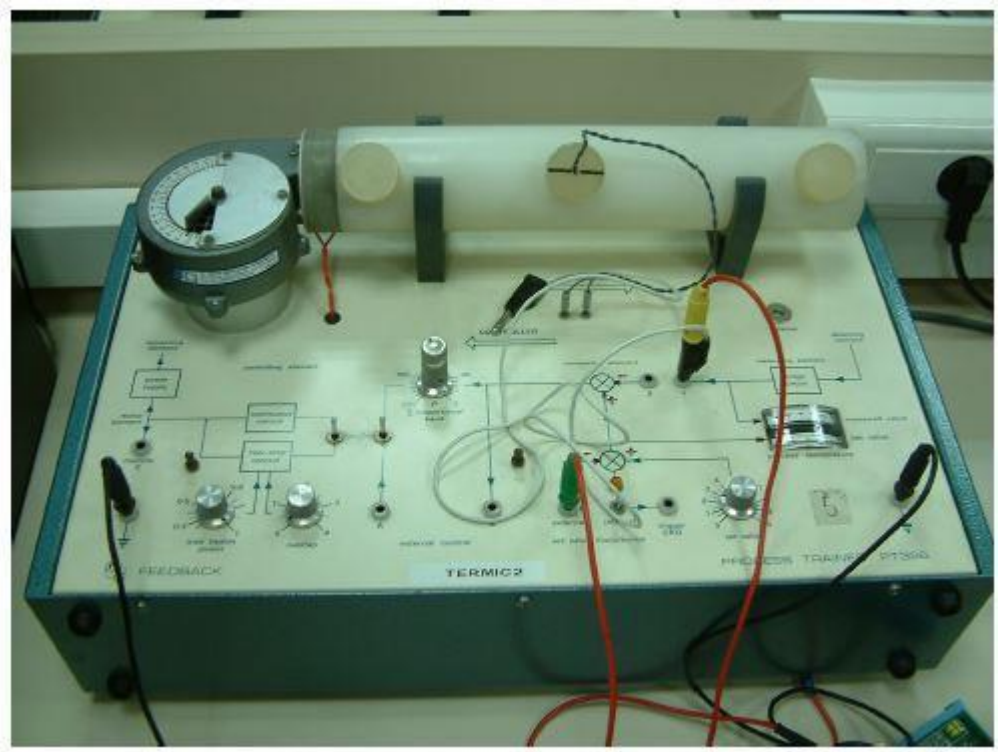


Figure 1: PT326 process (Picture Extracted from http://arvc.umh.es/ceaRsc/doc/pt326.jpg)

The process works as follows: air circulates through a tube and is heated in the inlet. The air temperature is measured with a sensor (thermistor). The input of the system is a voltage applied to the heating device, which is a mesh of resistor wires. The output of the process is the outlet air temperature represented by the measured thermistor voltage amplified by the electronics of the device.

The project is divided into two main parts:

i) The first part was pre-processing the data and identifying the system using different ARX structures to obtain the best result.

ii) The second part consisted of developing a three-block and a deadbeat controller and evaluating the capacity of the system to follow a reference and maintain the correct output in presence of perturbations.

PART I

2. Data collection and analysis

We processed three packages of data containing input-output, corresponding to three acquisition trials obtained in the same conditions. The acquisition board (National Instruments, NI-6009) can send the voltage to the system in the range 0V-5V and read the voltage from the system in the range of ±10V.

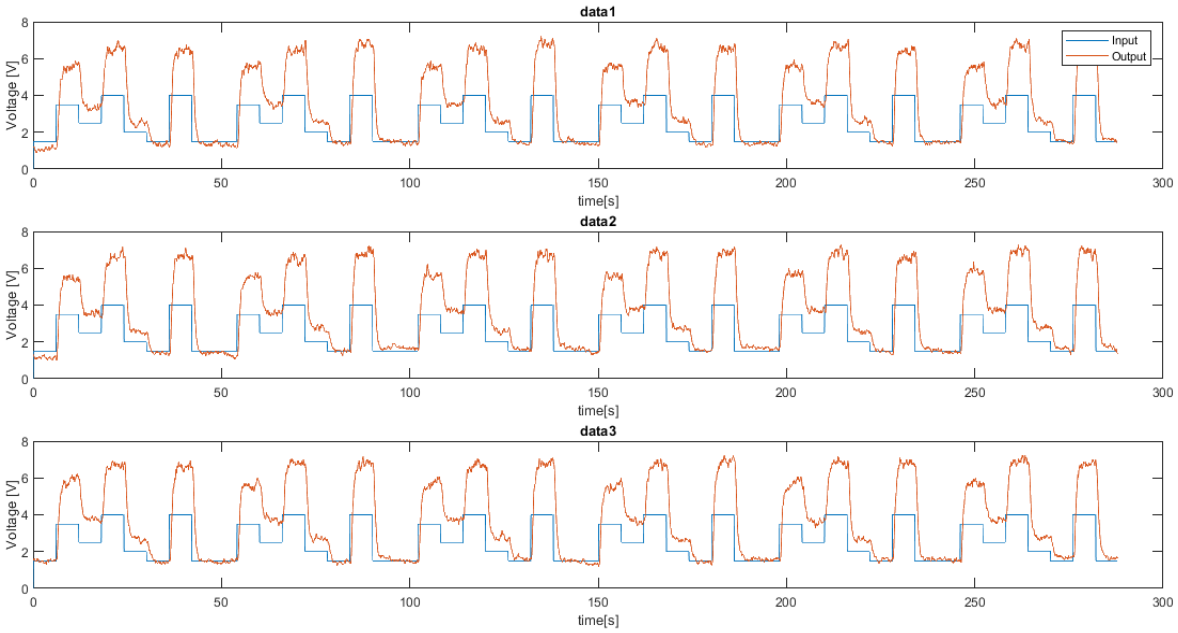


Figure 2: input-output of three acquisition trials obtained in the same conditions

As shown in figure 2 the blue signals (input) do not exceed 5V while the red one (output) oscillates in the range 0V-10V.

To convert the data from Volts to Celsius degrees we used the formula *T= 2.865V + 26.323*. The inverse formula *V=0.349T-9.20* can obviously be used to convert from Celsius to Volt.

3. Identification

3.1 Pre-processing

We used the function *detred* for every package of data to remove 1st and 2nd order trends, which correspond respectively to linear and quadratic mean trends.
The first two samples were removed from all the signals, because transients related to the beginning of the acquisitions may occur. Then we plotted the data again.

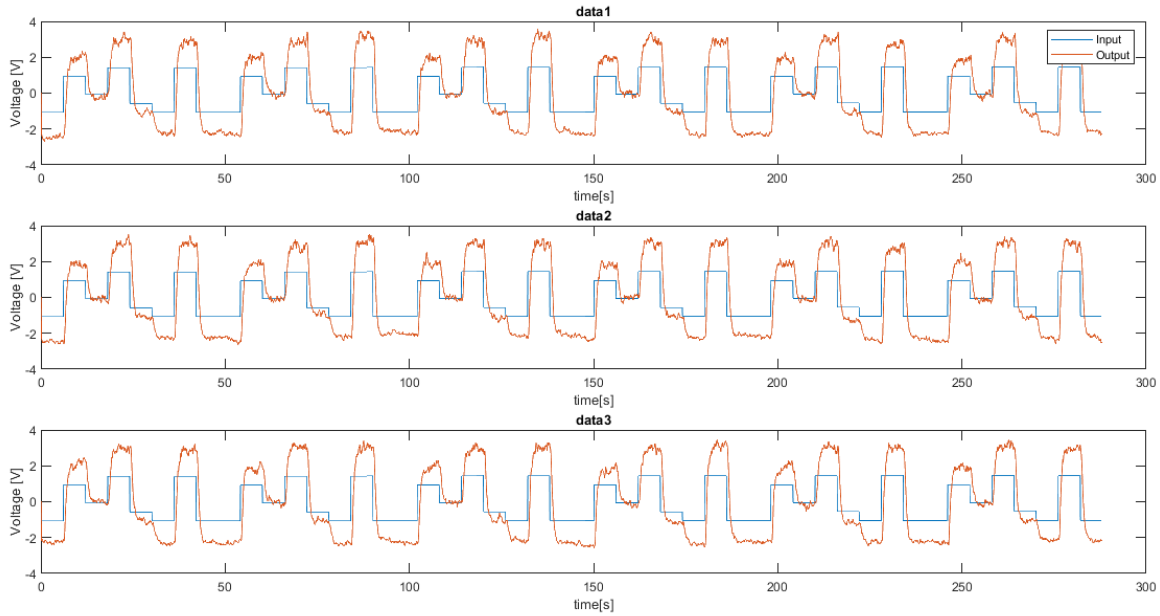


Figure 3: Data without 1st and 2nd order trends

3.2 ARX analysis

In order to find an appropriate identification of the system we evaluated different models of the ARX family by varying the number of poles (0:20), zeros (0:20), and delay (0:10).

For this purpose, the *arxstruct* function was applied to all possible useful combinations between the pre-processed data. The best fit is evaluated with two different criteria that look for the best fit penalizing solutions that use too many parameters :

i) Akaike Information Criterion (AIC)
ii) Rissanen's Minimum Description Length (MDL)*

*the mdl model is not required for the assignment, but analyzing it made the view of the arx structure wider.

The models found were then compared using all data through the *compare* function.

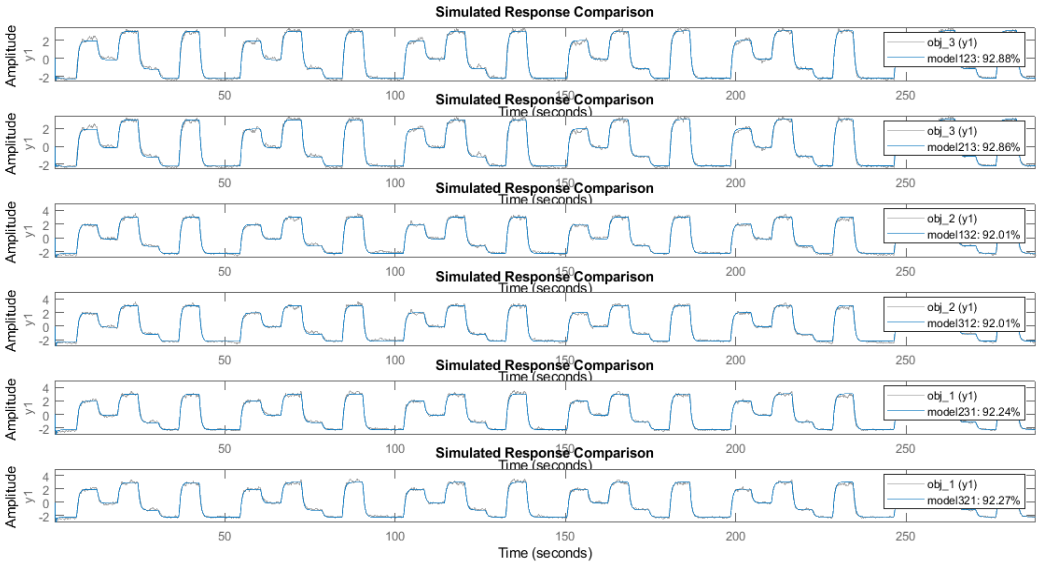


Figure 4: Models obtained using AIC criteria

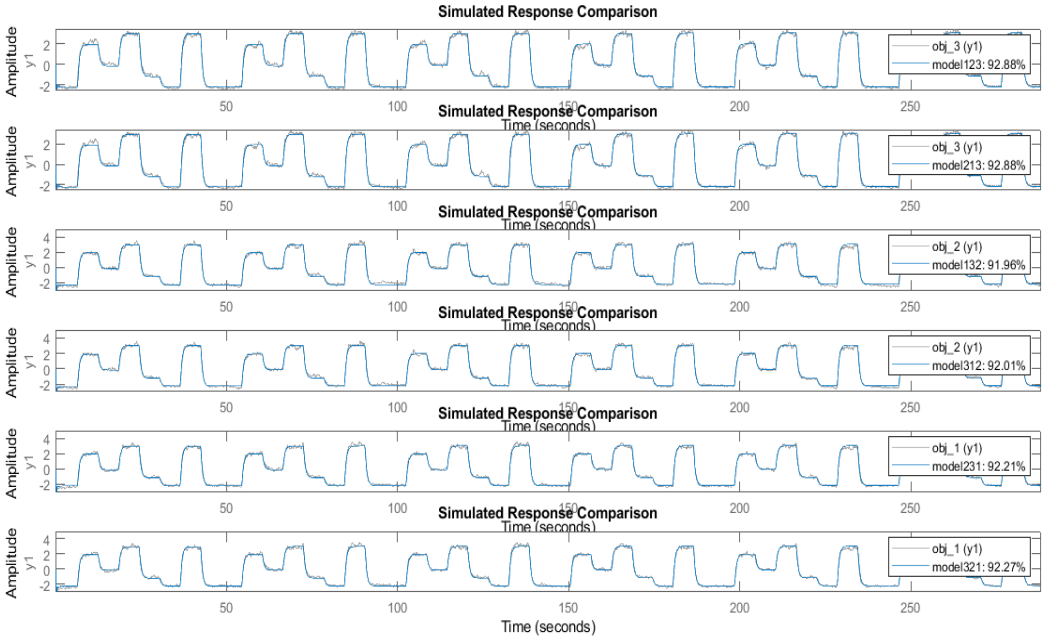


Figure 5: Models obtained using MDL criteria

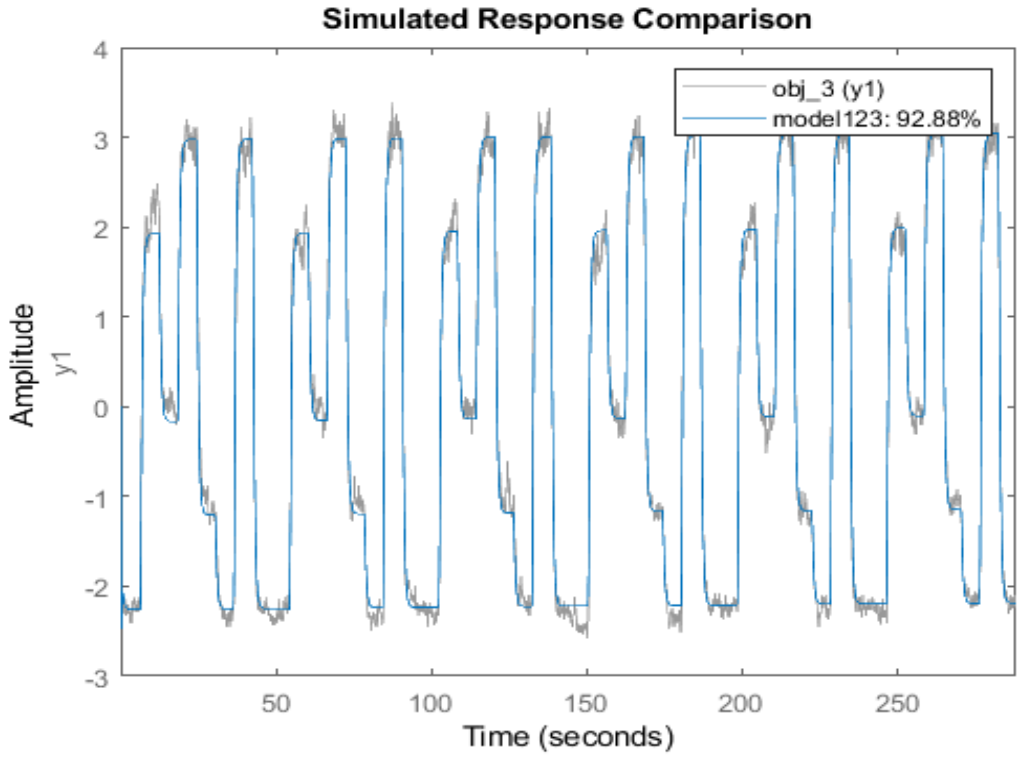


Figure 6: best fit obtained by AIC criterion evaluating the data_3 on the data_2 : number of poles=1, zeros=2 and delay=3

The results are summarized in the following Table

	EVALUATED DATA	COMPARED DATE	BEST FIT		POLO	ZERO	DELAY
			ICA	MDL			
	2	3	92.88%	92.88%	1	2	3
DATA 3	1	3	92.88%	92.86%	2	1	3
	3	2	92.01%	91.96%	1	3	2
DATA 2	1	2	92.01%	92.01%	3	1	2
	3	1	92.23%	92.21%	2	3	1
DATA 1	2	1	92.27%	92.27%	3	2	1

Table 1:The table above shows the best fit for both patterns, with relevant setup details. The table analyses all the possible combinations between the data.

We also explored the *arx structure* that produce the best fit using the app *system identification* of MATLAB and we estimated the system using a polynomial model, which varies the number of poles (0:20), zeros (0:20) and the delay (0:10).

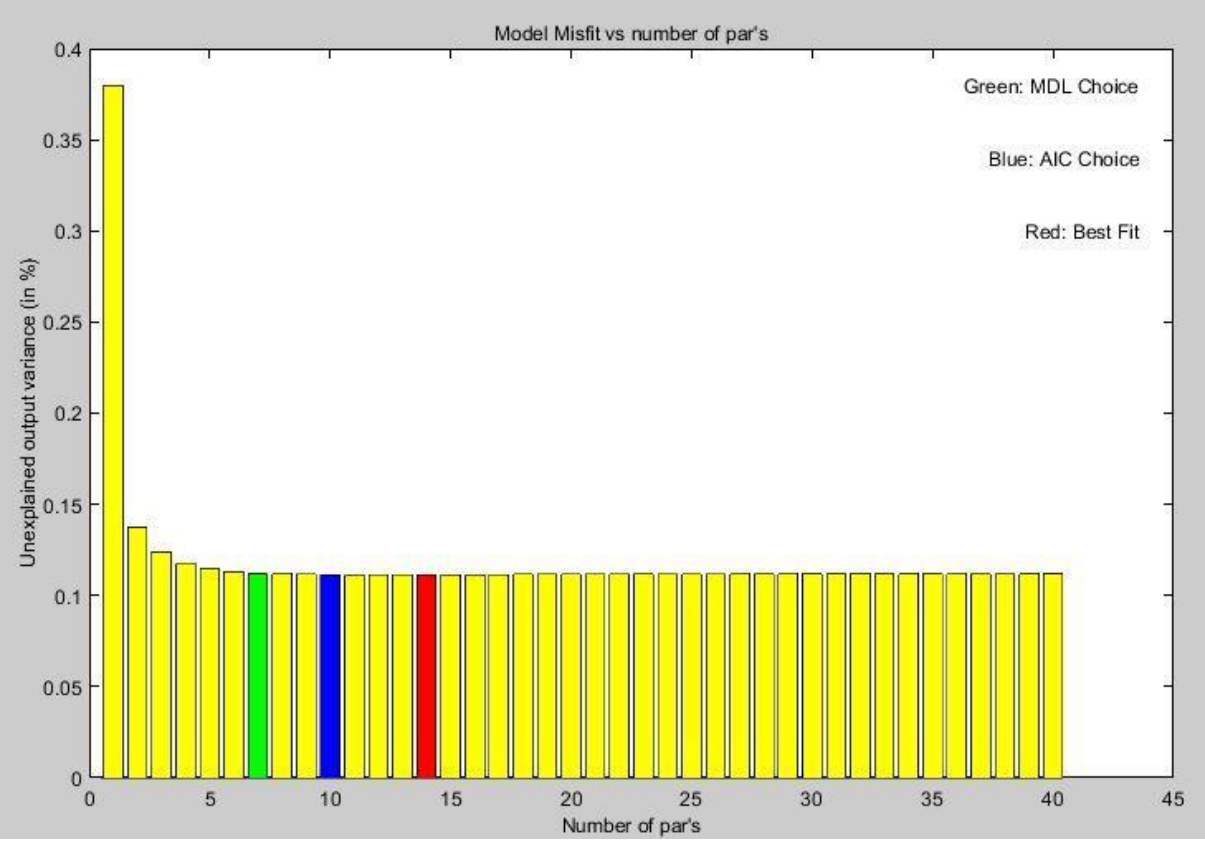


Figure 8: Unexplained output Variance of the arx pattern - data_3 - system identification

To conclude this part, the difference equation and the discrete-time ARX model of the best system found using AIC criterion are summarized below (data 3):

- Difference equation

Knowing that the transfer function is $G(z) = B(z)/A(z)$ and that the Z transform of x $[n - n_0]$ is $x[z]^{-n_0}$, we can write the difference equation as:

$$A(z) = 1 - 0.888 \cdot z^{-1}$$

$$B(z) = 0.05651 \cdot z^{-3} + 0.183 \cdot z^{-4}$$

(1-0.888 $\delta[x-1]$) × y[n] = (0.05651 $\delta[x-3]$ + 0.183 $\delta[x-4]$) × x[n]

- Discrete-time ARX model

A(z)y(t) = B(z)u(t) + e(t) with sample time= 0.08 sec

$$A(z) = 1 - 0.888\, z^{-1}$$

$$B(z) = 0.05651 \cdot z^{-3} + 0.183 \cdot z^{-4}$$

Polynomial orders: na=1 nb=2 nk=3

PART II

In order to do this part we considered an ARX structure with 2 poles, 2 zeros and a time delay of 3 samples (na=2, nb=2, nk=3). We proceeded on calculating the chosen ARX model which is given by the expression $A(z)y(t) = B(z)u(t) + e(t)$, where the polynomes A(z) and B(z) are written below.

$$A(z) = 1 - 0.8356z^{-1} - 0.04653z^{-2}$$
$$B(z) = 0.05794z^{-3} + 0.1937z^{-4}$$

1. Discrete controllers development and Simulink implementation

In this point we developed both a deadbeat controller and a three block controller considering the discrete time transfer function as PT326.

Controllers are used in closed loop to command other systems. The controller receives as input the error, which is the difference between the input and the output obtained by negative feedback and aims at decreasing this error in each control loop.

1.1 Deadbeat

Deadbeat is a type of controller that quickly reaches zero error at the sampling instants (output in a steady-state) and has little ripples between samples. The general scheme of a deadbeat controller can be seen in the figure below.

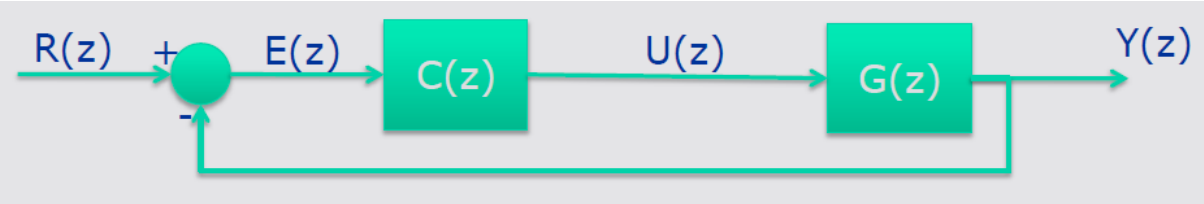


Figure 9: Deadbeat controller diagram

Using A(z) and B(z) calculated at the previous step we can write $G(z) = \frac{B(z)}{A(z)}$

And using the formula for deadbeat controllers $C(z) = \frac{A(z)}{B(1)-B(z)}$ we obtain the transfer function of the controller.

We then developed the simulink model represented below, in order to evaluate the controller.

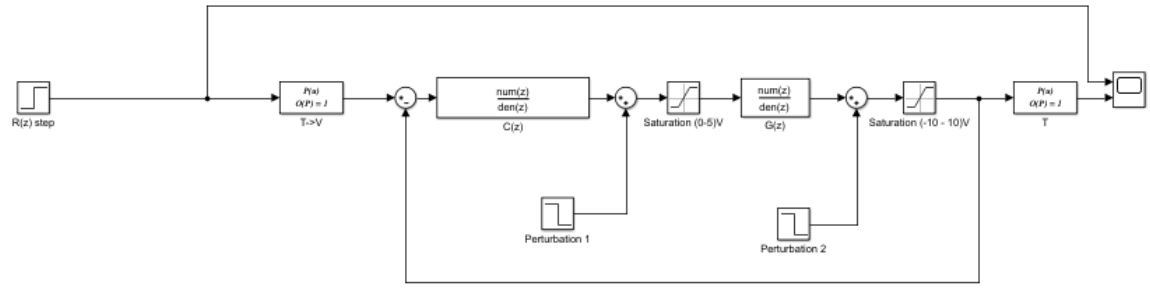


Figure 10: Simulink model of Deadbeat controller, using discrete transfer function.

In order to evaluate the performance of the controller we tested it using a square wave, step and a stairstep as the reference. At this point we don't use any perturbation.

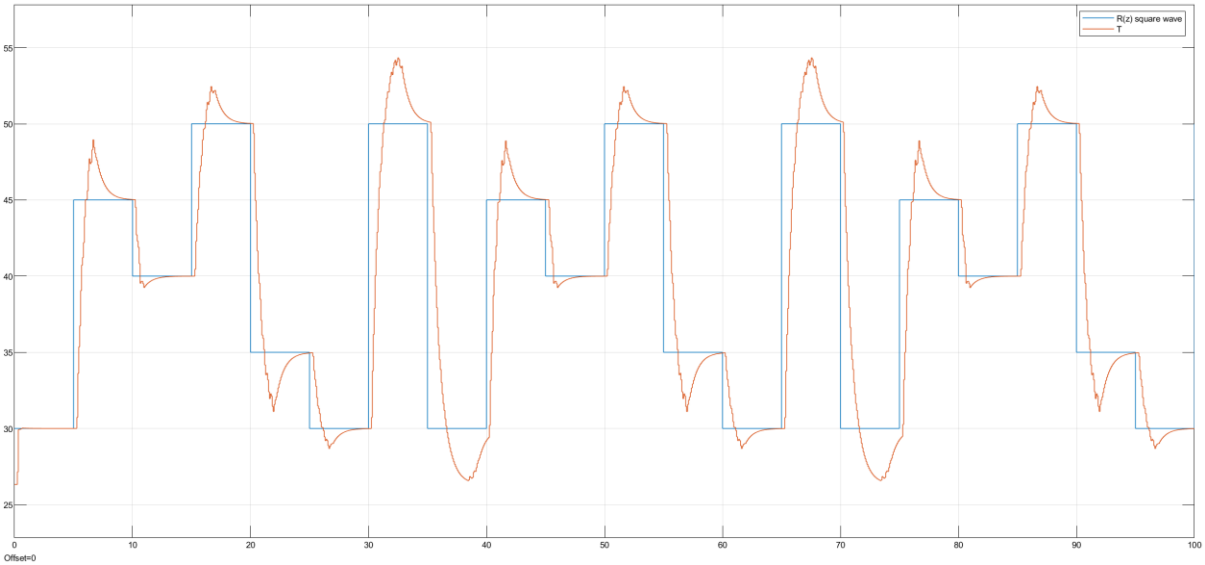


Figure 11: Results of Deadbeat controller following a square wave reference, using discrete transfer function.

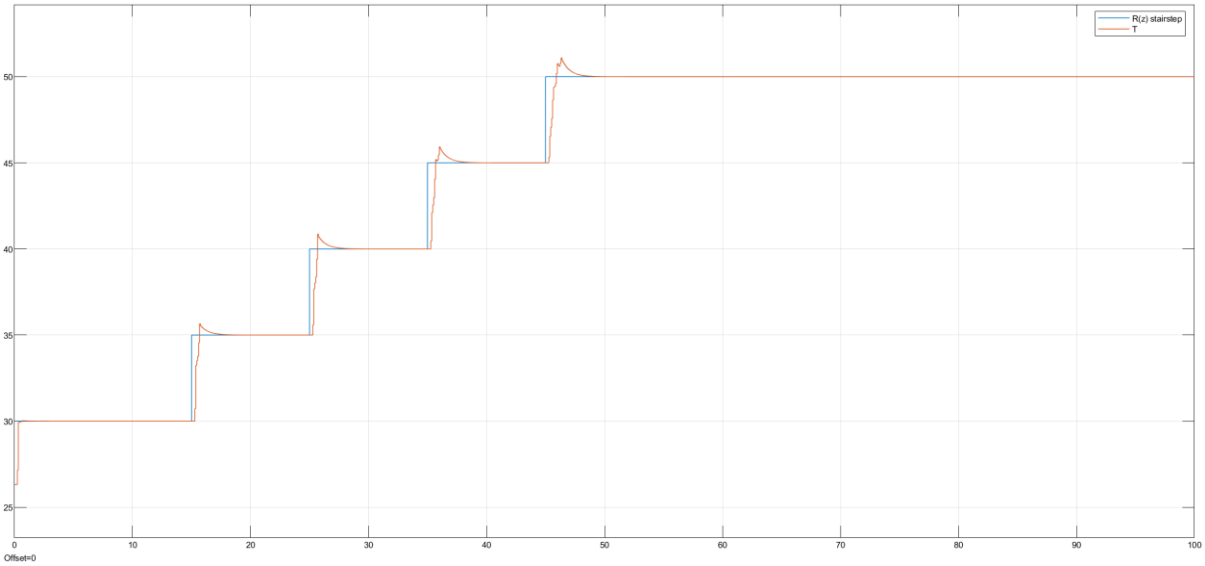


Figure 12: Results of Deadbeat controller following a stairstep reference, using discrete transfer function.

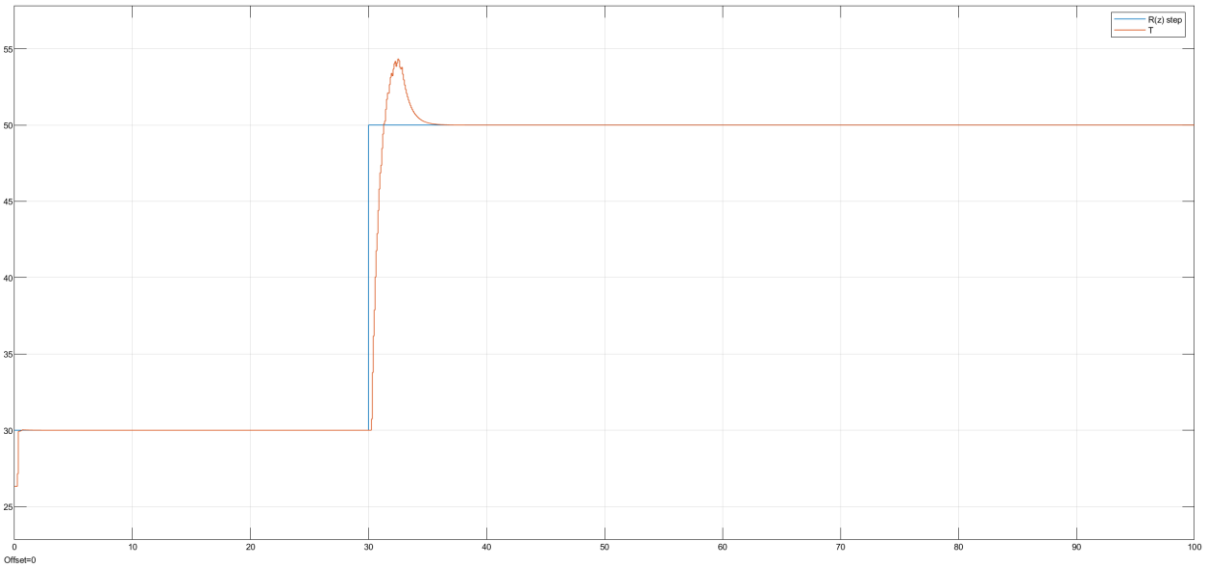


Figure 13: Results of Deadbeat controller following a step reference, using discrete transfer function.

We conclude that the controller is able to follow the reference even though it sometimes exceeds the reference.

1.2 Three-block controller

A more flexible controller is given by the Three Blocks Controller, which has the following general structure:

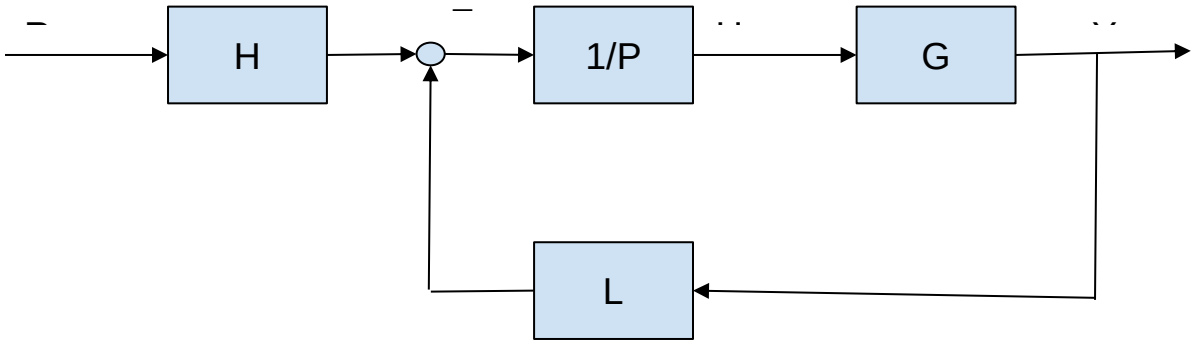


Figure 14: Threeblock controller diagram

To calculate the polynomial P, L and the gain H we solved the Diophantine equation:

$$P \cdot A + B \cdot L = T$$

Where the polynomial T is provided by the text of the assignment and is

$$T = (1 - 0.42 \cdot z^{-1}) \cdot (1 - 0.72 \cdot z^{-1})$$

A and *B* have been calculated at the previous step and are

$$A(z) = 1 - 0.8356 \cdot z^{-1} - 0.04653 \cdot z^{-2}$$
$$B(z) = 0.05794 \cdot z^{-3} + 0.1937 \cdot z^{-4}$$

The problem is now finding the parameters of P and L. The solution is unique if the number of parameters respects the following equations:

$$np = nb - 1$$
$$nl = na - 1$$
$$nt <= na + nb - 1$$

The parameters *nb*, *na* and *nt* are given by the order of the polynomials *A(z)*, *B(z)* and *T(z)* and are *nb*=4, *na*=2 and *nt*=2. We obtain that *np*=3 and *nl*=1.

We can now write the equation $P \cdot A + B \cdot L = T$ as

$$(1 + p1 \cdot z^{-1} + p2 \cdot z^{-2} + p3 \cdot z^{-3}) \cdot (1 - 0.8356 \cdot z^{-1} - 0.04653 \cdot z^{-2}) +$$
$$+(0.05794 \cdot z^{-3} + 0.1937 \cdot z^{-4}) \cdot (l0 + l1 \cdot z^{-1}) = (1 - 0.42 \cdot z^{-1}) \cdot (1 - 0.72 \cdot z^{-1})$$

Solving the equation with MATLAB we obtain the values of *p1, p2, p3, l0* and *l1* and therefore the polynomials P and L.

$$P(z) = (1 - 0.3044 \cdot z^{-1} + 0.0946 \cdot z^{-2} + 0.0510 \cdot z^{-3})$$
$$L(z) = (0.2391 + \ 0.0123 \cdot z^{-1})$$

The gain H can be calculated solving

$$H(1) = \frac{T(1)}{B(1)}$$

We implemented the system in Simulink with the following configuration without noise sources and using a step input:

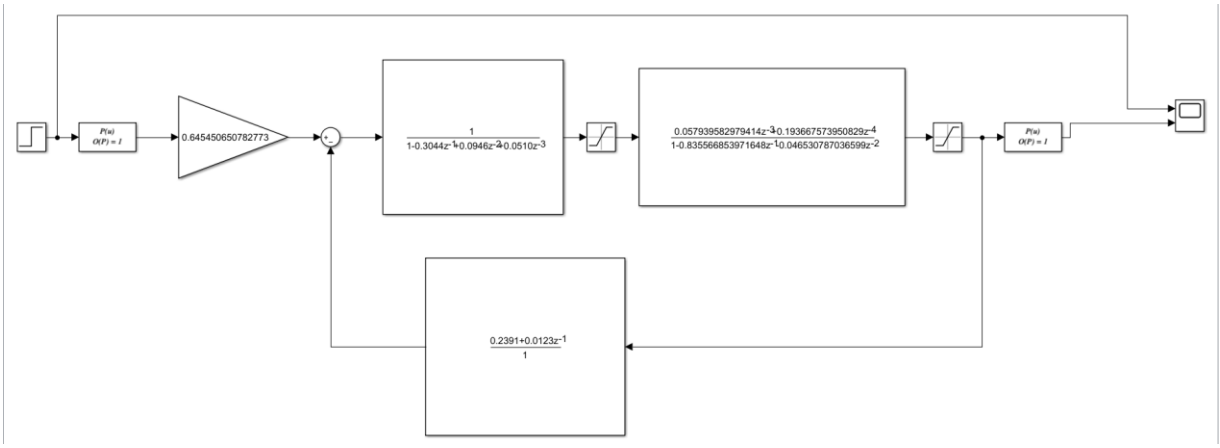


Figure 15: Simulink model of Three-block controller, using discrete transfer function.

We obtained the following output result:

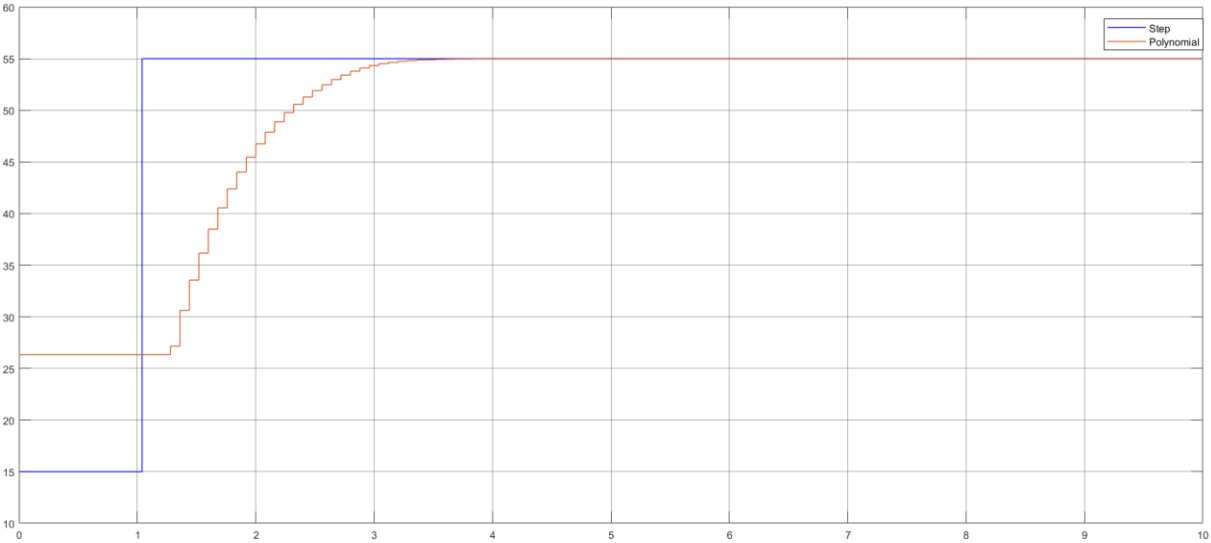


Figure 16: *Results of Three Blocks controller following a step reference, using discrete transfer function.*

We also tried different inputs to see how well the system can follow the reference. In some cases the effect of the saturation can be clearly seen: the system cannot go lower than 26°C. In general the performance is satisfactory, with no ripples or overshooting.

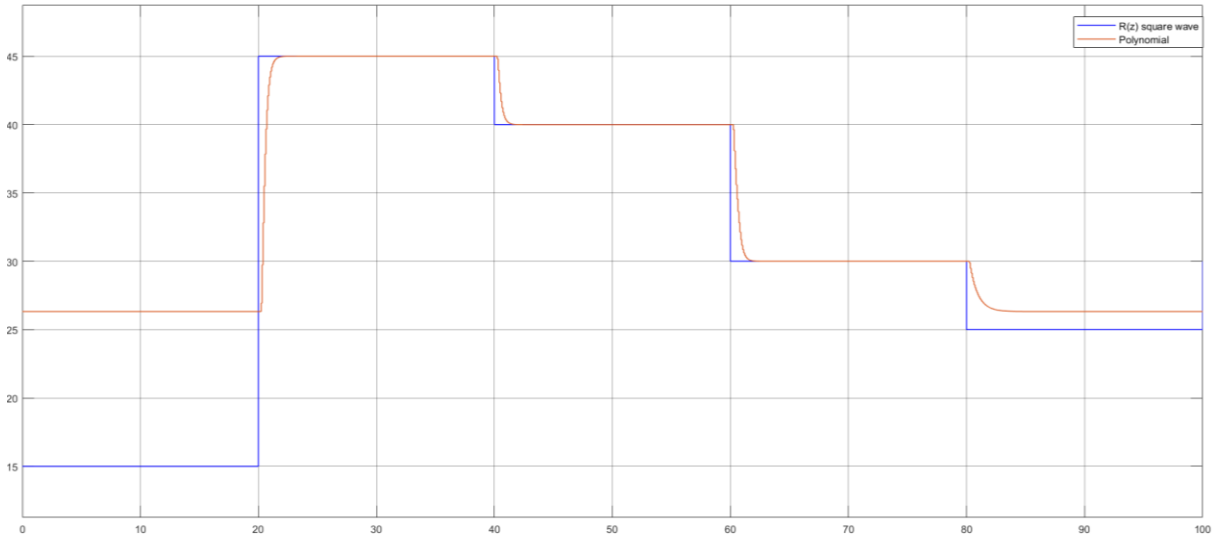


Figure 17: Result output of Three Blocks controller following a square wave reference, using discrete transfer function.

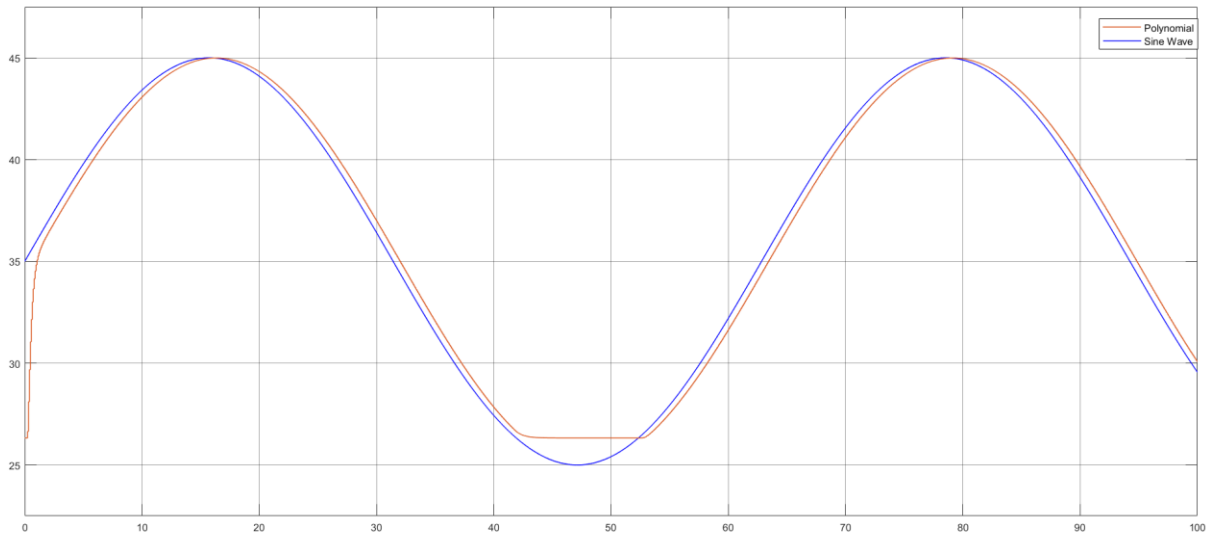


Figure 18: Result output of Three Blocks controller following a sine wave reference, using discrete transfer function.

It can also be clearly seen the delay in following the reference.

2. Application to continuous model and Simulink implementation

In this point we evaluated the controllers considering that PT326 is given by a continuous transfer function G(s).

$$G(s) = e^{-0.2s}\frac{1.05s + 18.14}{s^2 + 5.98s + 8.69}$$

The continuous transfer function is a better approximation of the real system that we want to control.

2.1 Deadbeat

In this point we tested the performance using the same references of point 1.1 and didn't use any perturbation.

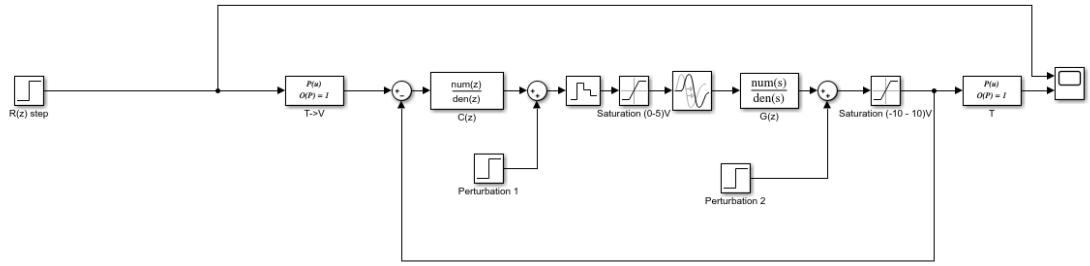


Figure 19: Simulink model of Deadbeat controller, using continuous transfer function.

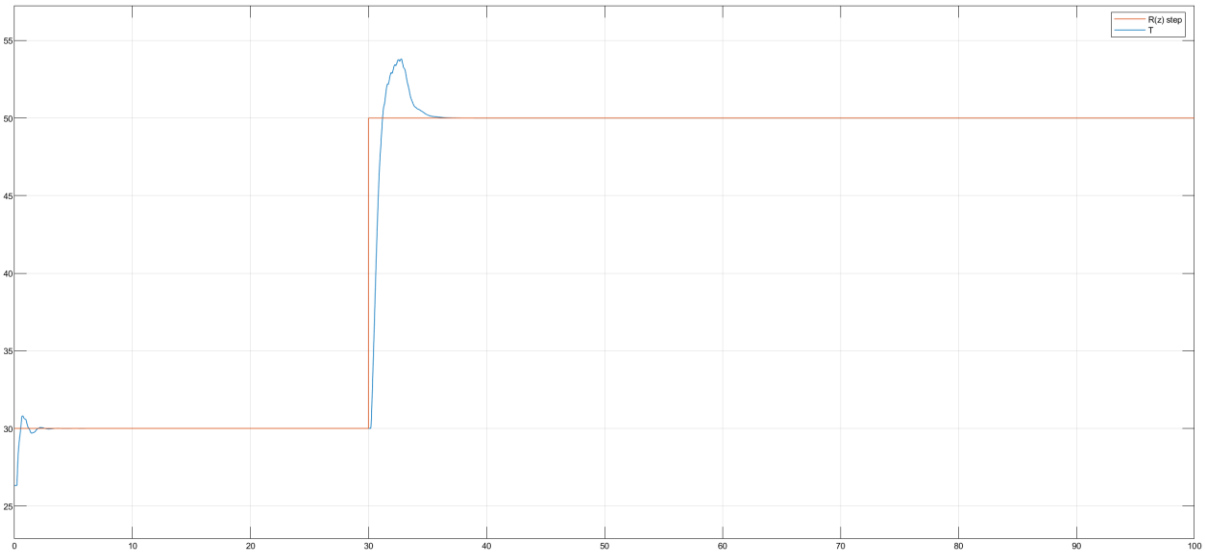


Figure 20: Results of Deadbeat controller following a step reference, using continuous transfer function.

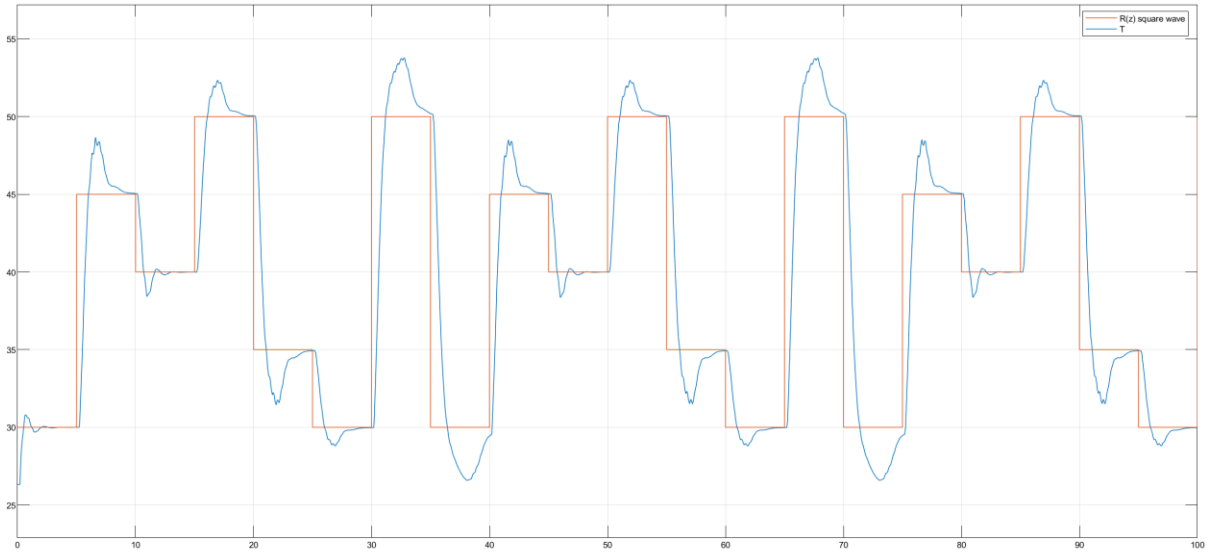


Figure 21: Results of Deadbeat controller following a square wave reference, using continuous transfer function.

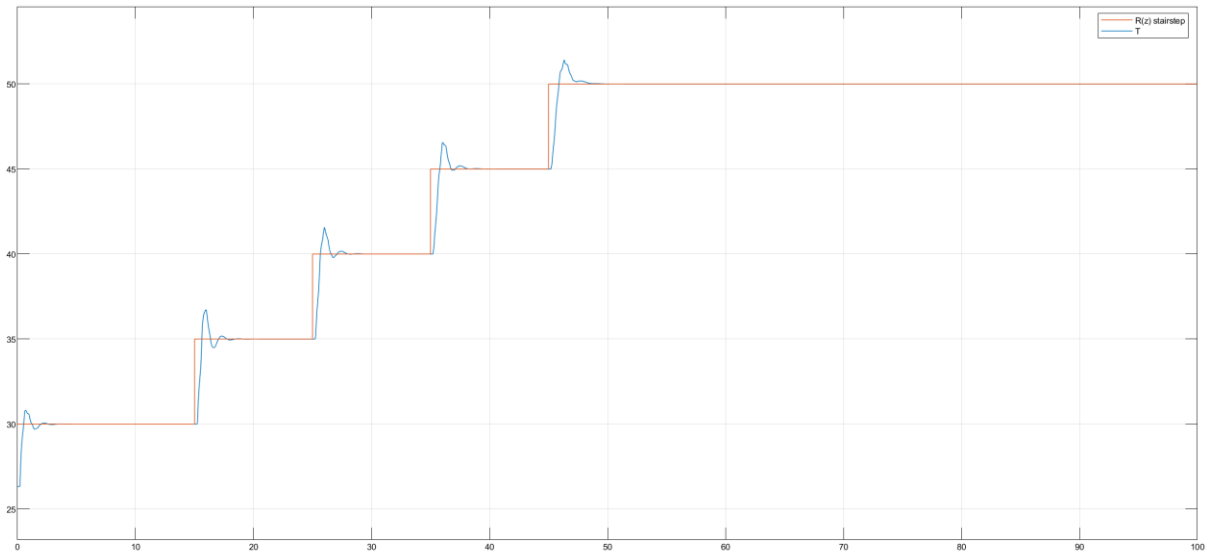


Figure 22: Results of Deadbeat controller following a stairstep wave reference, using discrete transfer function.

We conclude that the controller is able to follow the reference also with the continuous time transfer function G(s).

2.2 Three blocks

In this point we tested the three blocks controller. PT326 is given by the continuous transfer function G(s).

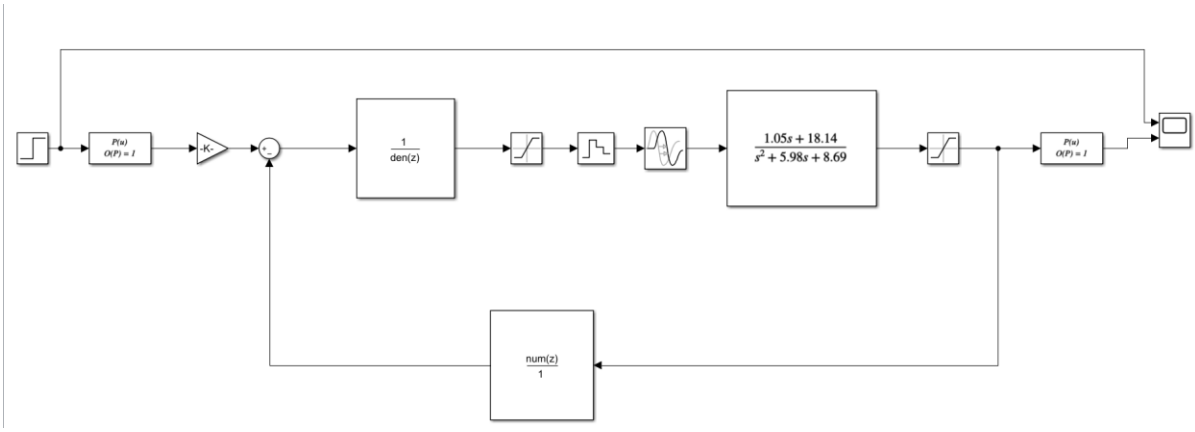


Figure 23: Simulink model of Three-block controller, using continuous transfer function.

As already done in the discrete case we tested the system with different inputs.

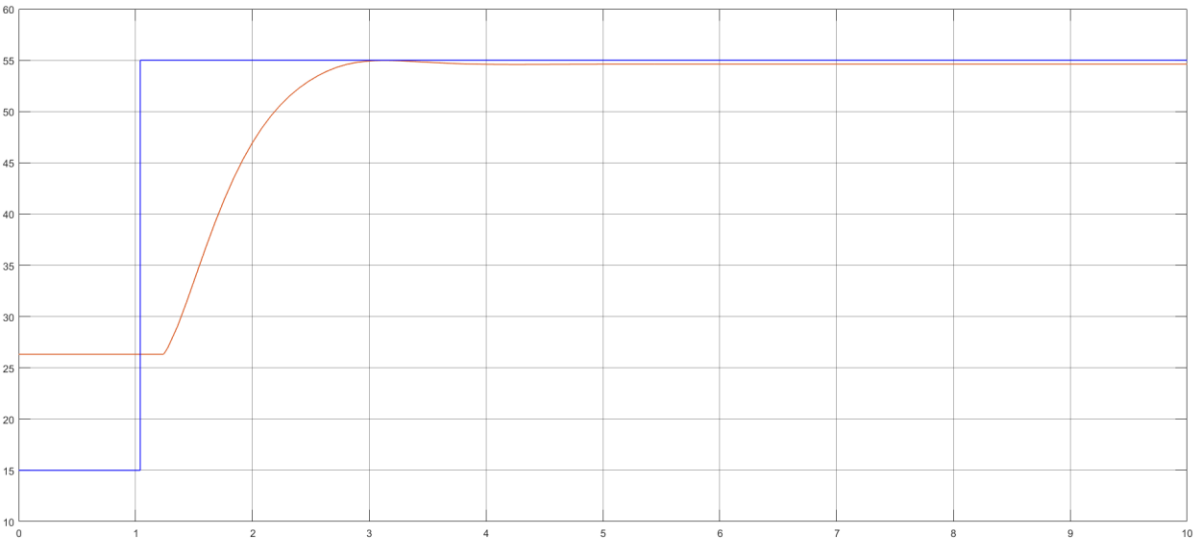


Figure 24: Results of Three Blocks controller following a step reference, using continuous transfer function.

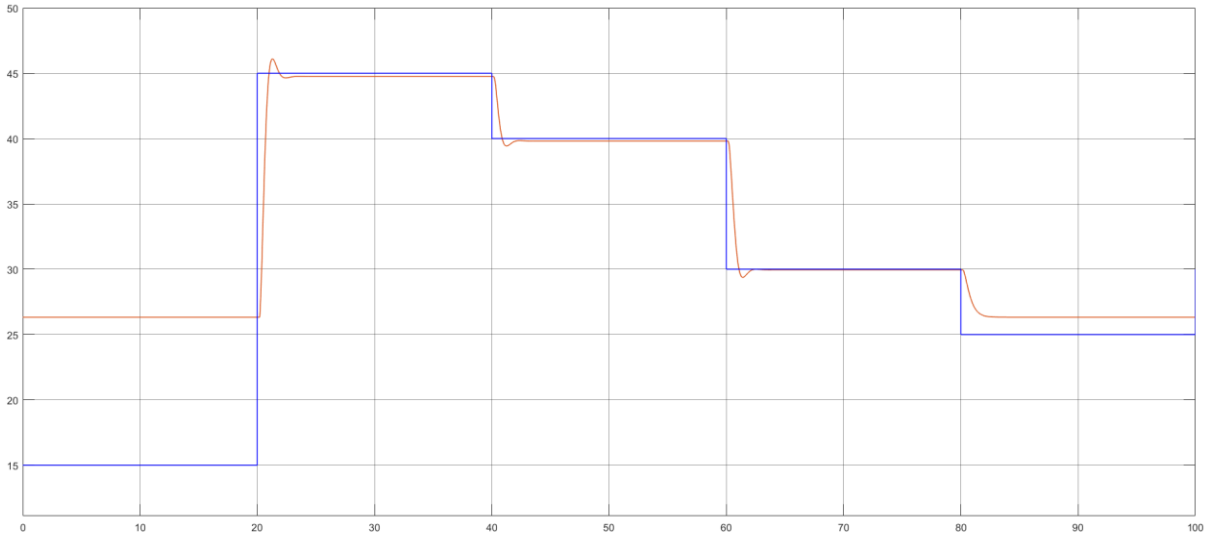


Figure 25: Results of Three Blocks controller following a square wave reference, using continuous transfer function.

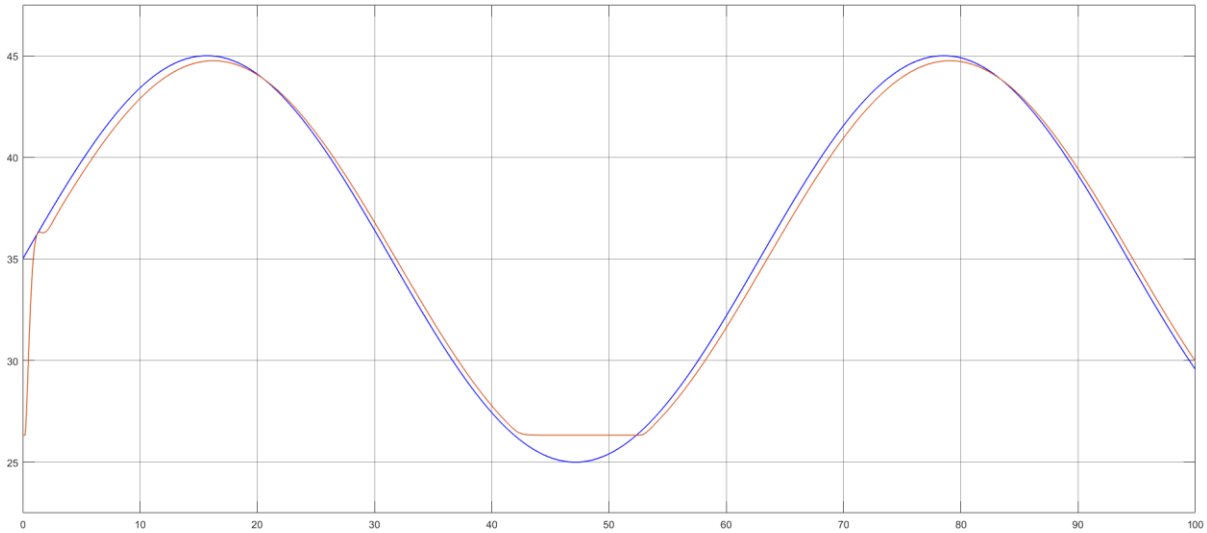


Figure 26: Results of Three Blocks controller following a sine wave reference, using continuous transfer function.

We can notice that the system never reaches exactly the reference. We hypothesize that this problem might be caused by errors of approximation or by the fact that the discrete transfer function $\frac{B(z)}{A(z)}$ used to calculate the controller slightly differs from the continuous function $G(s)$provided in the assignment.

4. Test of the systems against perturbations.

In this point we evaluated the capacity of the controllers to overcome perturbations. The first perturbation is the actuator and the second is the load disturbance.

4.1.1 Deadbeat perturbations using discrete transfer function.

Both the input and output perturbations are ±5V.

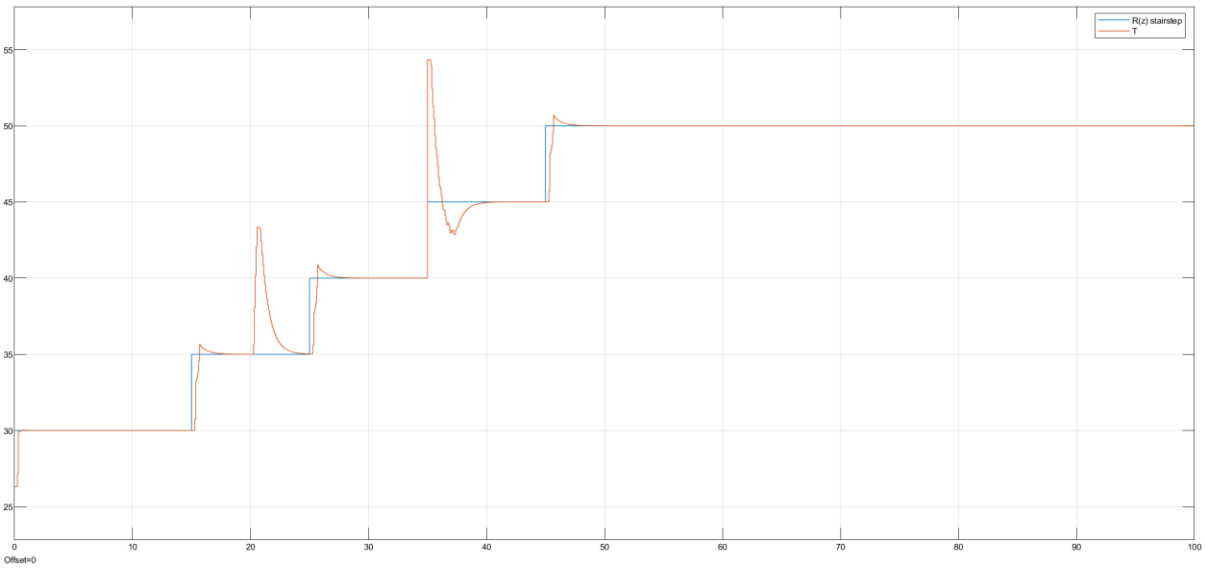


Figure 27: Output of the discrete system controlled using a deadbeat controller disturbed with two step noises with positive sign, applied after 20 and 35 seconds.

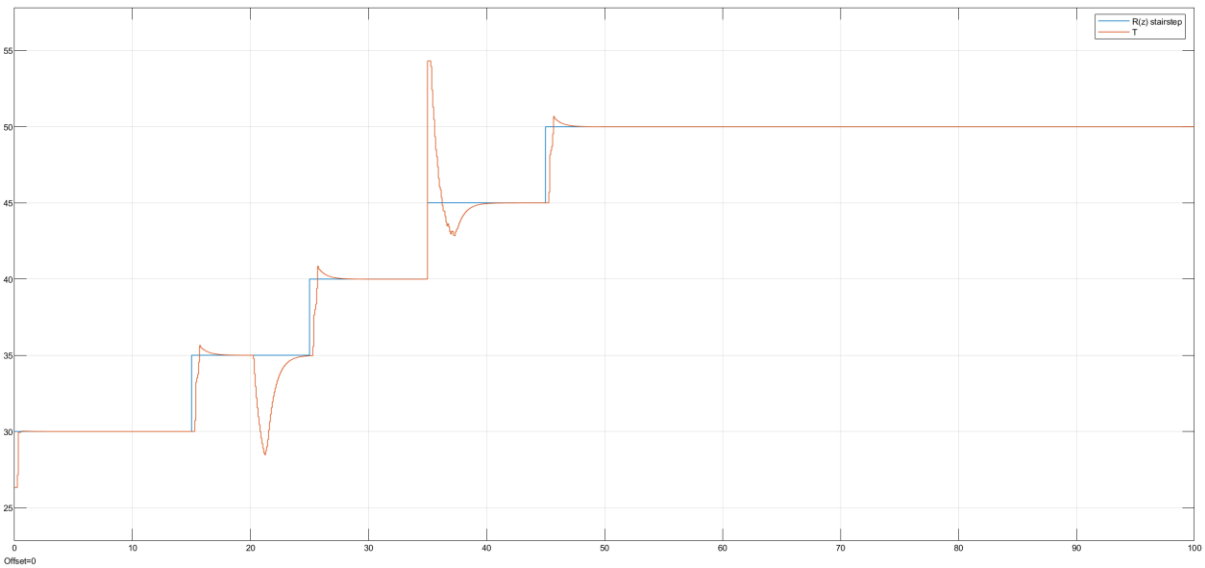


Figure 28: Output of the discrete system controlled using a deadbeat controller disturbed with two step noises with negative and positive sign, applied after 20 and 35 seconds.

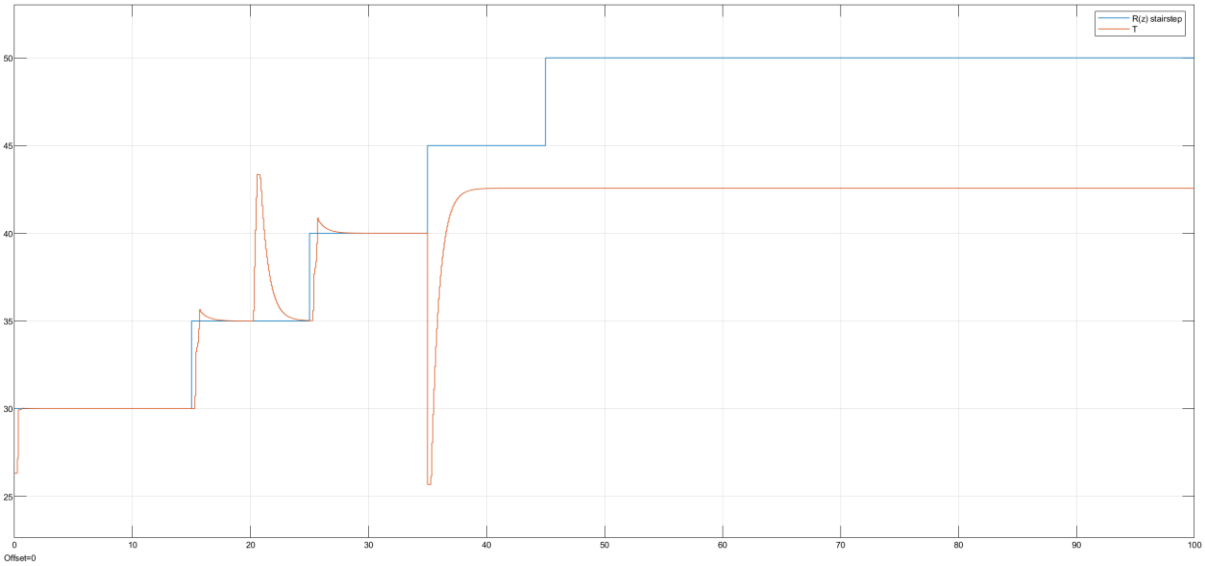


Figure 29: Output of the discrete system controlled using a deadbeat controller disturbed with two step noises with positive and negative sign, applied after 20 and 35 seconds.

4.1.2 Deadbeat perturbations using continuous transfer function.

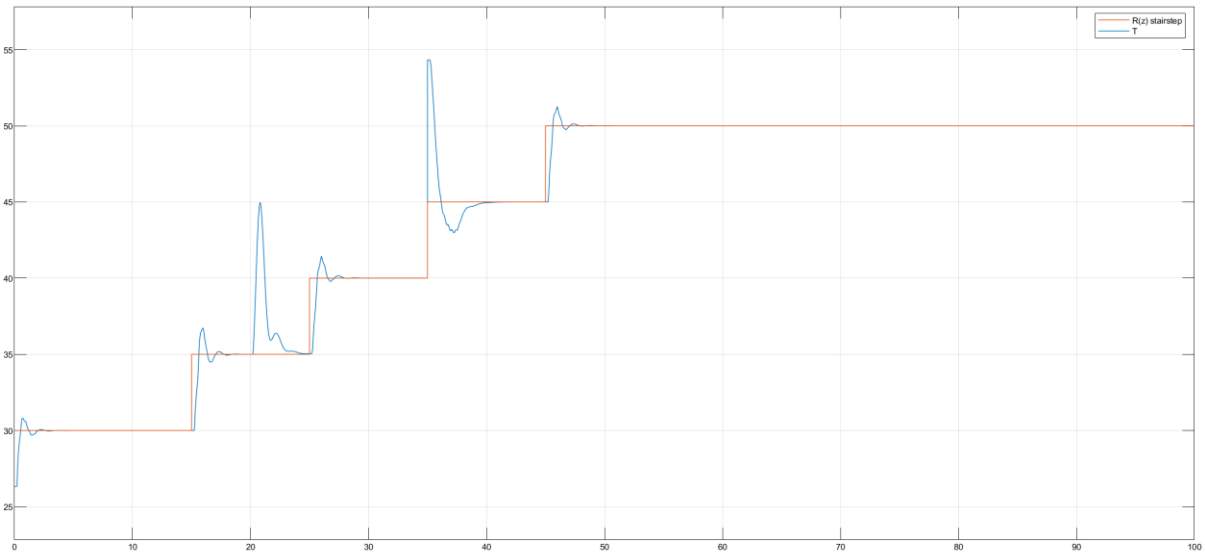


Figure 30: Output of the discrete system controlled using a deadbeat controller disturbed with two step noises with positive sign, applied after 20 and 35 seconds.

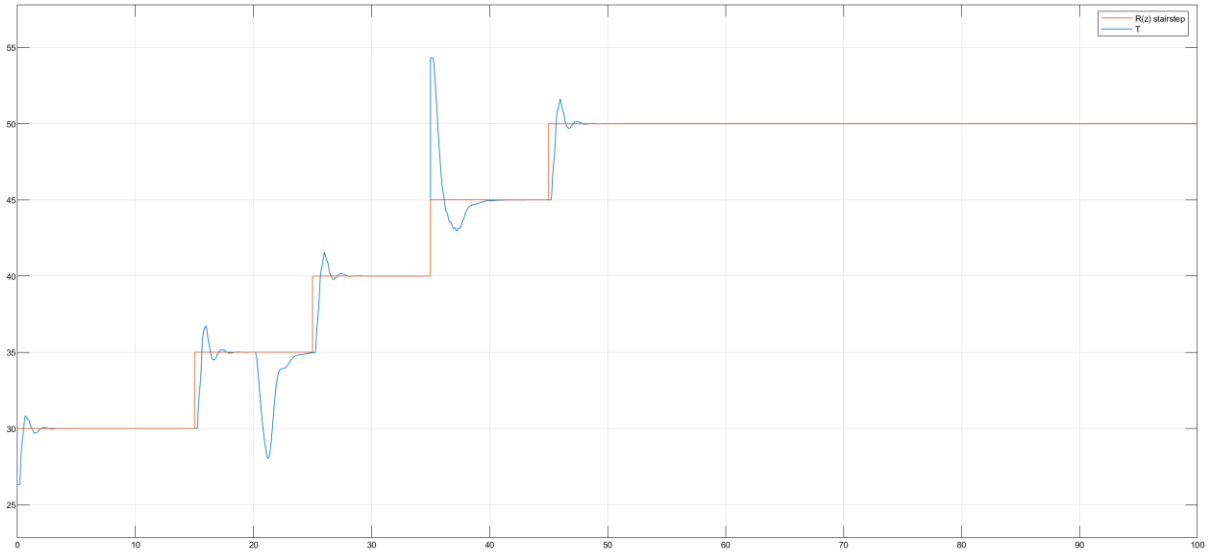


Figure 31: Output of the discrete system controlled using a deadbeat controller disturbed with two step noises with negative and positive sign, applied after 20 and 35 seconds.

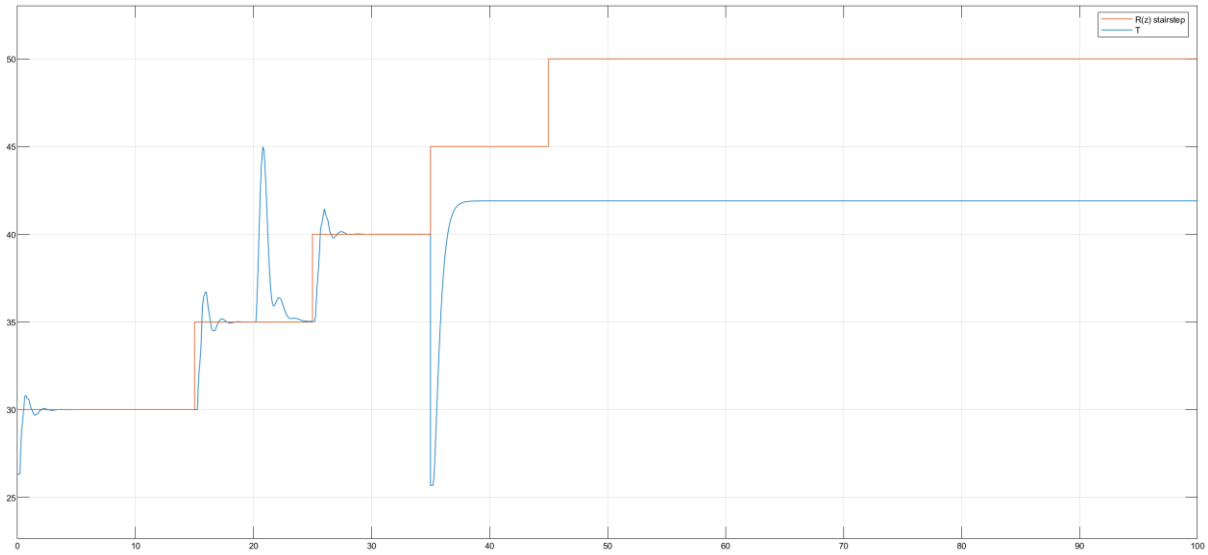


Figure 32: Output of the discrete system controlled using a deadbeat controller disturbed with two step noises with positive and negative sign, applied after 20 and 35 seconds.

4.2.1 Three blocks perturbations using discrete transfer function.

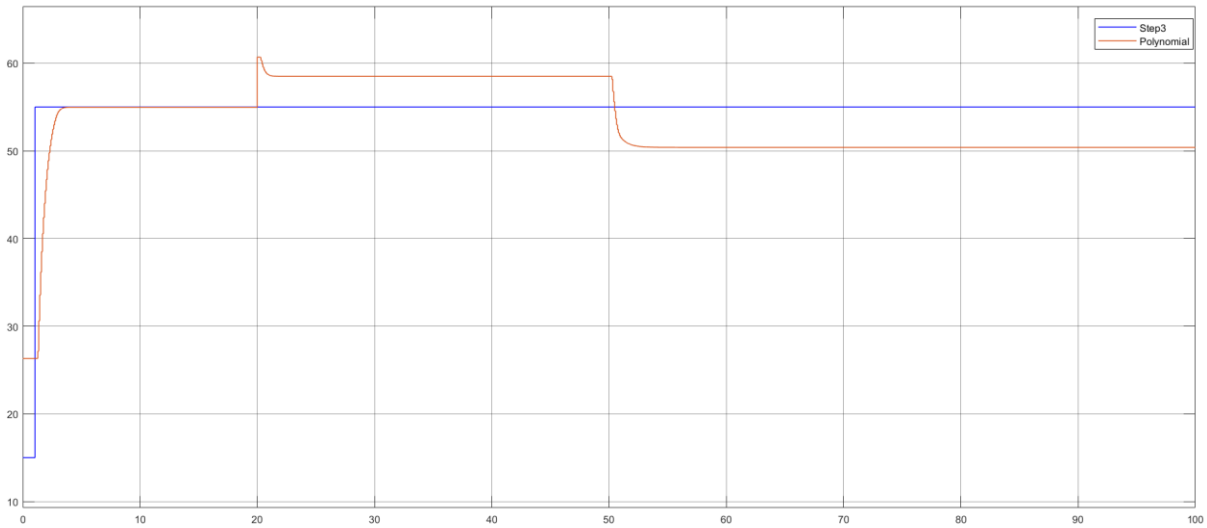


Figure 33: Output of the discrete system controlled using a three blocks controller disturbed with two step noises with opposite signs applied after 20 and 50 seconds.

The output of the system has the same shape of the reference but remains biased when we add load and/or actuator disturbances, even using a smaller disturbance.

4.2.2 Three block perturbations using continuous transfer function.

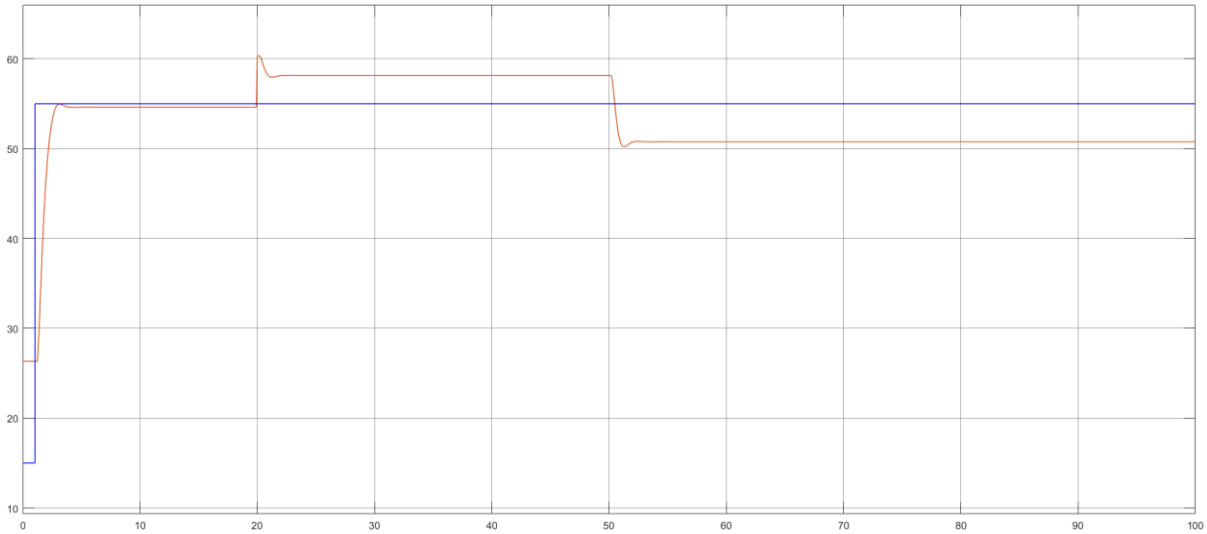


Figure 34: Output of the continuous system controlled using a three blocks controller disturbed with two step noises with opposite signs applied after 20 and 50 seconds.

The result is very similar to the one obtained in the discrete case. In both cases the system can follow the reference when the noise has zero mean, as in the case of white noise.

5. Capacity of the control system to follow the reference

We now want to measure how well the output can follow the reference when the discrete system is controlled by the deadbeat and the three block controller. The input reference used is a sinusoid. The actuator and the load disturbances are white noises.

Deadbeat controller:

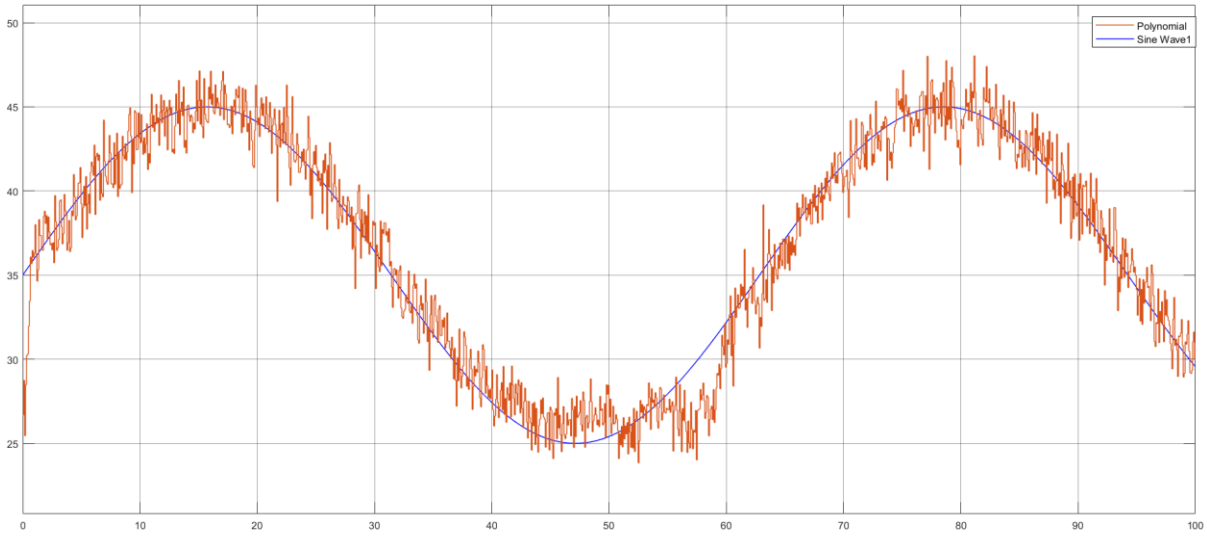


Figure 35: Reaction to white noise of the Deadbeat controller.

Three blocks controller:

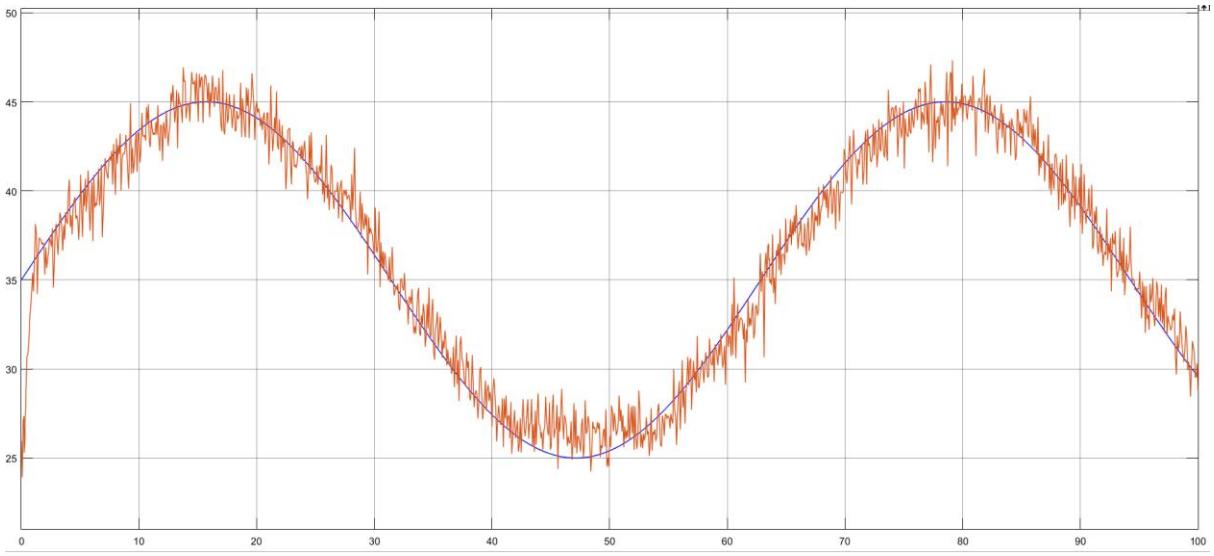


Figure 36: Reaction to white noise of the Three-blocks controller.

The root mean square error (RMSE) between the reference and the output is 1.5363 for the deadbeat controller and 1.4025 for the three block controller.

In this configuration we can conclude that the three blocks controller follows better the reference. This is not true if a constant noise is applied since the three blocks controller remains biased.

6. Results

Regarding the deadbeat we can conclude that the controller works properly, being able the system to follow the reference. The results obtained using the discrete and the continuous transfer functions are similar even though we can see the downgrade using the continuous. This is expected since it is more similar to a real scenario.

The deadbeat controller can adjust to any perturbation in the input but, if the output perturbation is too high, the error will increase a lot, requiring the system to reach a higher voltage. If the voltage required to follow the reference is higher than 5V, since the system only works between 0-5V, the controller stops working properly being unable to follow it. This happens for the discrete and the continuous transfer functions.

The three blocks controller can quickly reach the reference when tested with different inputs and when disturbed using white noise but cannot follow the reference when a constant disturbance is applied. For this reason we believe that the deadbeat controller should be preferred.